Orthopolity: a general law of resource allocation

A natural-law proposal, physical realizations and cosmic implications

Research manuscript, revised 11 November 2026. Detailed derivations, complete comparisons and analysis histories are retained in the supporting material. Empirical studies include exploration, locally frozen forecasts and retrospective reanalyses; none is externally preregistered. Authorship and submission declarations require agreement before submission.

Abstract

We propose orthopolity as a natural law of resource allocation: nature tends to distribute resources equally among their concentrations, relative to a physically specified comparison measure. The idea is simple: if each larger object contains more resource, fewer such objects are needed to hold the same class total. Abundance times mean resource per object is therefore constant in a neutral allocation. We distinguish this physical proposal from exact accounting and show how exchange dynamics, radiative transport and thermal modes realize specified equalities. Logarithmic and linear size classes give different predictions; independently specified constraints predict departures. Restricted combinations of resources also allow predictions to transfer between environments. Retained tests include accurate engineered allocation, chemical and language transfers, biological budget closure, and failed forecasts that constrain the explanation. These establish conditional results, not a common neutral regime across all natural populations examined. A supporting cosmological conjecture asks when cosmic assembly produces equal mass per logarithmic mass interval. Orthopolity organizes these mechanisms and interpretations around one question: what physically determines how much resource each class receives?

1. The law in one picture

We propose orthopolity as a natural law: nature tends to distribute resources equally among concentrations of those resources, relative to a physically specified comparison measure. This develops the proposal of Fabbri and Oliveira Jr. (2017) and Fabbri (2024). Its general physical reach is proposed; the exact results below hold under their stated conditions.

Consider objects of sizes $k = 1, 2, 4, 8$, with resource requirement $q = k$. Give each of these four classes 8 units. The classes can then contain 8, 4, 2 and 1 objects, respectively. The same total appears in four different arrangements:

$$8 \times 1 = 4 \times 2 = 2 \times 4 = 1 \times 8 = 8.$$

There are fewer expensive objects, but each class holds the same resource. Figure 1 makes the distinction visible: the blocks differ; the stacks have identical heights. Here the four equally weighted classes form a logarithmic size grid, so the example is **log-orthopolic**. The total budget is 32.

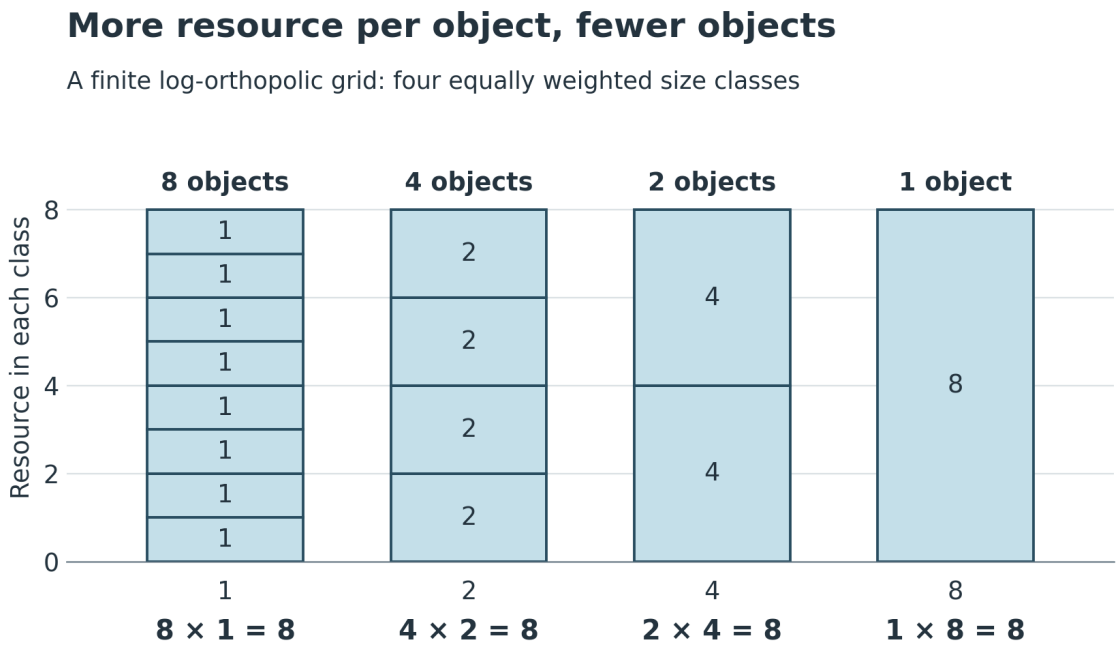


Figure 1. Count times resource per object gives resource per class. Each outlined block is one object; its height is its resource requirement. The four geometric classes have counts 8, 4, 2 and 1 and costs 1, 2, 4 and 8, respectively. Each class holds 8 units. This prescribed finite log-orthopolic allocation illustrates the relation; it is not an observation or a continuous-bin approximation.

2. From the picture to a prediction

2.1 The allocation relation

For any class, the starting equation is ordinary accounting:

$$\text{number of objects} \times \text{mean resource per object} = \text{resource in the class.}$$

To compare a continuous range of sizes k, let μ specify the class measure, $n_\mu = dN/d\mu$ the number density and $\bar{q}(k) > 0$ the arithmetic mean resource per object of size k. The resource density is

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k) n_\mu(k). \tag{1}$$

A **neutral orthopolic allocation** has equal resource per unit of that measure:

$$\mathcal{O}_\mu(k) = C > 0, \qquad n_\mu(k) = \frac{C}{\bar{q}(k)}. \tag{2}$$

Thus doubling the mean cost halves abundance per equal measure. If a domain has positive finite measure $\mu(D)$ and resource budget B, then $C = B/\mu(D)$. Equation (1) accounts for resource; equation (2) adds the allocation claim.

One sufficient allocation condition is symmetry among equally weighted classes. If their shares are exchangeable and total resource is fixed, each class has the same expected share. Fixed positive costs then give inverse-cost expected counts. Such a symmetry must be physically justified. Equal expected shares also allow unequal snapshots, as when the entire budget occupies a randomly chosen class. The observation must match the claimed average.

2.2 The measure changes the exponent

The measure answers a concrete question: what makes two classes equally wide? The intervals 1–2, 2–4 and 4–8 have equal logarithmic width, whereas 1–2, 2–3 and 3–4 have equal linear width. They make different allocation comparisons.

Write $p_\mu = n_\mu/N$ for a normalized object density over a finite population. If cost is $\bar{q}(k) = q_0(k/k_0)^\alpha$, with fixed positive reference values q_0, k_0, equation (2) becomes

$$(k/k_0)^\alpha p_\mu(k) = c, \qquad c = \frac{C}{N q_0}.$$

This is the familiar $k^\alpha p(k) =$ constant relation when size is expressed in units of k_0: an increasing cost and a decreasing frequency exactly compensate. Its meaning depends on what density p denotes. Let p_{lin} be density per dk and $p_{\log}$ density per $d\ln k$. They obey $p_{\log} = k p_{\text{lin}}$, so

Allocation hypothesis	**Constant resource density divided by N**	**Predicted density per linear size**
Linear-orthopolic: equal resource per dk	$\bar{q}p_{\text{lin}}$	$p_{\text{lin}} \propto k^{-\alpha}$
Log-orthopolic: equal resource per $d\ln k$	$\bar{q}p_{\log} = \bar{q}k p_{\text{lin}}$	$p_{\text{lin}} \propto k^{-(\alpha+1)}$

For example, if the resource per object is its mass, equal mass per logarithmic mass interval predicts density proportional to mass to the power -2. Equal mass per linear mass interval

predicts power -1. Neither measure follows from conservation alone. Choosing a measure after seeing the distribution can manufacture equality; a physical prediction fixes it independently.

The inverse-cost relation also applies to costs that are not powers. We call its normalized profile a *natural (orthopolic) distribution*. Power laws are one member of this resource-based family, not its definition.

2.3 What accounting makes unavoidable

For a nonnegative additive resource, a budget B can support at most B/t disjoint objects each requiring at least $t > 0$ units. If a complete partition has N objects with mean resource $\bar{q}$, then $N\bar{q} = B$ exactly. This remains true at every complete level of a nested hierarchy, even when the objects within a level differ in size.

These levels repeatedly describe the same whole. Coexisting disjoint classes instead divide it: class j holds $N_j\bar{q}_j = Bf_j$, with $\sum_j f_j \leq 1$. Equal shares require an additional condition on the fractions f_j. The finite-capacity bound and complete-partition identity are mathematically unavoidable; equal allocation among competing classes is a further physical statement. Supporting S1.1 and S1.4 derive coverage and overlap corrections. This accounting result is neither a log-orthopolic nor a linear-orthopolic size-allocation test by itself.

2.4 Resources, constraints and a second prediction

A resource can combine several requirements. A useful explanation must specify how those requirements combine and how the environment affects allocation. Write the constrained relation as

$$\bar{q}_{\mathrm{eff}}(k)n_\mu(k) = Ca(k), \qquad a(k) > 0. \tag{3}$$

Here a represents independently determined opportunities or constraints. For example, absorption predicts how radiative power decreases with distance. An unrestricted a fitted to the observed profile explains nothing further. Likewise, choosing a cost proportional to $1/n_\mu$ always constructs neutrality. An exploratory resource reading becomes evidence through independently fixed inputs and a second prediction that passes.

Restricted resource families make such predictions possible. Suppose effective cost is $Q = Q_0 \prod_\ell X_\ell^{\theta_\ell}$, measured constituents scale as $X_{s\ell} \propto k^{D_{s\ell}}$ in environment s, and an independently known constraint is $a_s \propto k^{-v_s}$. For log-orthopolic allocation, a linear-size count density $k^{-(1+\alpha_s)}$ then implies

$$b = D\theta, \qquad b_s = \alpha_s - v_s. \tag{4}$$

In words, independently measured changes in the requirements constrain how the abundance exponent can change. Two calibration environments with constituent rows $(1, 1)$ and $(1, 2)$ identify two shared exponents; a third row $(2, 1)$ predicts $b_3 = 3b_1 - b_2$. The third abundance slope is a separate target.

This assumes deterministic class requirements or suitable size-independent residual moments. Nonlinear combinations also need a fixed physical partition; they need not preserve resource under subdivision. Supporting S1.2–S1.5 retain the feasibility, averaging, identification and uncertainty results, including synthetic checks. The contribution follows established inverse-problem methods (Keshavarz et al., 2011; Habeck, 2014) and depends on the physical inputs and their successful transfer.

3. Ideal mechanisms and physical realizations

The orthopolic umbrella connects several kinds of explanation. Complete accounting establishes how counts and mean resource relate across descriptions. An allocation mechanism explains how class shares are selected. A constrained realization predicts how a specified process changes those shares. An exploratory interpretation proposes a resource reading that still needs an independent test. Their mathematical compatibility gives a common language; their evidential roles remain different.

The following realizations show what each explanation actually equalizes. Ecological size-spectrum studies provide a direct precedent for comparing biomass across size classes (Sheldon et al., 1972; Gaedke, 1993; Hatton et al., 2021). Detailed proofs and complete comparisons remain in supporting material.

3.1 Equalization, a constraint and its removal

Resource flows from classes with higher resource per unit reference width to classes with lower resource per width. When every class is connected, this exchange selects the neutral allocation.

Consider fixed classes with positive reference widths w_i, fixed costs q_i, and resource stocks R_i totaling B. Symmetric conductances $g_{ij} = g_{ji} \geq 0$ on a connected graph give the exchange current

$$J_{ij}^0 = g_{ij}\left(\frac{R_i}{w_i} - \frac{R_j}{w_j}\right).$$

The resulting dynamics conserve resource and relax to $R_i^0 = Bw_i/\sum_j w_j$, hence $N_i^0 = R_i^0/q_i$. This mechanism selects equal resource per the declared measure. It establishes the neutral relation within its specified model class.

A fixed class preference V_i changes the transition rate from i to j to $(g_{ij}/w_i)\exp[\max(V_i - V_j, 0)]$. Detailed balance then gives

$$R_i^V = \frac{Bw_i e^{-V_i}}{\sum_j w_j e^{-V_j}}.$$

The added rates balance the neutral current at constrained equilibrium. Removing the preference preserves the current state and predicts relaxation toward the neutral allocation. The generator's spectral gap supplies a convergence bound; disconnected components equalize only within their separately conserved budgets. These are established reversible-chain tools applied to resource classes (Levin and Peres, 2017).

Whole-object assembly gives asymptotic inverse-cost mean counts under symmetric saturated channels, with finite-budget corrections; mass-action kinetics on the same conserved budget instead favor larger objects in the worked case (supporting S3.10). Hidden link storage and nonproportional storage curves can also alter relaxation or intervention outcomes despite matched baseline measurements (supporting S5.19 and S3.7).

Passive RC networks cannot have positive imaginary impedance at their driven port. An independent forecast including magnetic storage gives 12.2% curve error, below a 39.4% lower bound for any passive RC network. Physical capacitance weights define this one-circuit test; the extension preserves energy accounting (supporting S5.20), without uniquely establishing the general law.

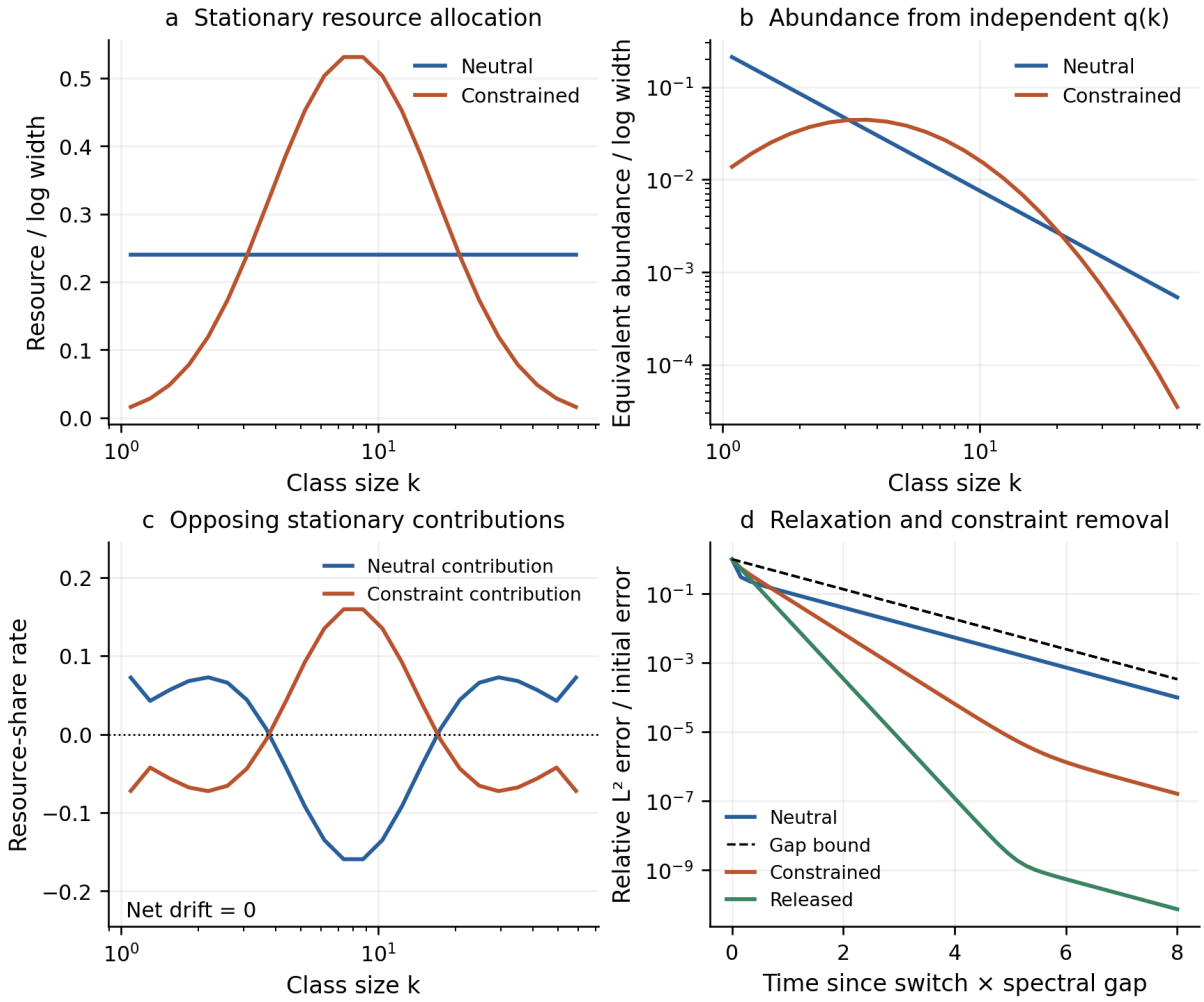


Figure 2. Resource exchange on fixed logarithmic size classes. The model predicts neutral allocation, a nonuniform constrained equilibrium and recovery after release. The detailed model, spectral bounds and finite-packet simulations are in the supporting material and model report.

3.2 Gravity and radiating waves: what geometry preserves

A larger enclosing surface can receive the same total flux while the flux per unit area falls. This is another inverse relation, but its objects and measure are surfaces and geometry rather than a size distribution of material objects.

For a Newtonian gravitational field $\mathbf{g}$, the Poisson equation (Chao, 2014) gives the integral relation $\oint_S \mathbf{g} \cdot \mathbf{n}\, dA = -4\pi G M_{\text{enc}}$. Every enclosing surface outside a fixed isolated source has the same inward field flux. Spherical symmetry additionally gives $|\mathbf{g}| = GM/r^2$: the conserved enclosed-source quantity is distributed over an area growing as r^2. Inward flux per solid angle is $d\Psi/d\Omega = GM$, an explicit angular equality. This is a geometric realization with specified symmetry. The field flux is not an energy throughput, and nested surfaces are not independent mass or energy shares. Distributed sources change $M_{\text{enc}}(r)$; nonspherical sources need not give uniform local fields. The gravity derivation states the resource interpretation, energy-accounting limits and departures.

For radiation, Poynting's balance gives a physical energy throughput. In steady transparent transport outside a source, the enclosing-surface power is $P(r) = L$; isotropic emission gives flux density $L/(4\pi r^2)$. With outgoing radial propagation at speed c, disjoint shell stocks instead satisfy $dU = (L/c)dr$. Equal radial thicknesses hold equal energy. Equal logarithmic radial intervals have $dU/d\ln r = Lr/c$, so they do not have equal stocks. The geometry identifies which comparison realizes equality.

An independently specified absorbing medium gives

$$P(r) = L \exp\left[-\int_{r_0}^{r} \kappa(s)\, ds\right].$$

It predicts the departure from constant radiative throughput and accounts for energy deposited in matter. If absorption is removed while the source continues, the response travels outward and the transparent profile returns by $(r - r_0)/c$. The transport derivation gives the full space-time response and its assumptions (Poynting, 1884; Haus and Melcher, 1989; Condon and Ransom, 2016). Neither the gravitational nor radiative inverse-square law alone determines the abundance of objects at different sizes.

3.3 Thermal modes: a derived limitation of neutral allocation

In classical equilibrium, each harmonic mode has the same mean energy. Quantum physics predicts where that equality breaks down.

A finite set of classical harmonic modes in canonical equilibrium has mean energy $k_B T$ per mode. Counting modes is the independently defined measure, neither log-orthopolic nor linear-orthopolic size allocation; instantaneous mode energies can fluctuate. For electromagnetic excitation energy above the ground state, quantization gives instead

$$\bar{\epsilon}_\nu = \frac{h\nu}{e^{h\nu/(k_B T)} - 1} = k_B T \frac{x}{e^x - 1}, \qquad x = \frac{h\nu}{k_B T}.$$

The classical plateau is approached for $x \ll 1$ and suppressed for $x \gg 1$. The departure follows from the allowed states before a spectrum is observed. Mode density $8\pi\nu^2/c^3$ then gives Planck's spectrum. Equal energy per mode is therefore different from equal energy per frequency interval (Planck, 1901; Tong, n.d.).

Our FIRAS calculation converts 43 microwave-background channels to excitation energy per mode. Their $x = 1.199$–11.262 range illustrates quantum suppression. The source already incorporates a fitted blackbody template and correlated errors (Fixsen et al., 1996); this is

established observational evidence, not new independent confirmation. Supporting material retains the derivation and spectrum.

For nonequilibrium cantilevers (Fontana and Bellon, 2023), colder-condition damping calibrates a quadratic local-loss model before hotter fluctuations are evaluated. Ten of eleven mode fits are physically admissible. On their 26 targets, mean absolute error is 74.0 K versus 78.9 K for uniform local loss. Large mode errors and one negative-dissipation fit show that global damping does not identify adequate spatial energy weights. Supporting S5.13 retains the retrospective design, calibration conventions, plotting omission and all failures. Inherited equilibrium normalization precludes a new equipartition test.

3.4 Cell growth and division: a measurable constraint envelope

Cells spend different amounts of time at different sizes. Birth and division therefore determine which sizes are available and how they are represented.

Let cell mass grow as $\dot{x} = gx$, with common deterministic g, conservative symmetric binary division and no death. For a balanced asynchronous population and stationary lineage of physiologically equivalent daughters, growth-fragmentation theory gives $p_L(x) = xp_P(x)/E_P[x]$ (Genthon, 2022). Birth mass B and division mass D satisfy $D \overset{d}{=} 2B$. The cycle-crossing fraction is

$$H(x) = P(B \leq x < D) = F_B(x) - F_B(x/2),$$

and the population law becomes

$$p_P(x) = \frac{2H(x)}{E[B^{-1}]x^2}, \qquad \frac{x^2 p_P(x)}{E_P[x]} = \frac{H(x)}{\ln 2}.$$

The right-hand side is normalized mass allocation per logarithmic size. Where every cycle crosses an interval, $H = 1$ and allocation is log-orthopolic. Fixed birth mass b gives $2b/x^2$ on $[b, 2b)$; variable births can remove the plateau. Independent birth observations predict the envelope. Growth noise, asymmetric division or lineage selection require a different observation model (derivation).

4. What the empirical comparisons establish

4.1 Predictions and comparisons

Forecasts fix objects, measure, costs, constraints and observation models. The registry and supporting material retain inputs, comparisons and prior access. Identities on the same data provide no independent confirmation; later freezing does not erase exposure.

Evidence	Established outcome	Scope or unresolved limitation
Runtime resource competition	Calibrated costs predict allocation and a thread-count intervention	One engineered mechanism
Size-selected algal cultures	Equal-biovolume forecasts transfer across selection treatments	Separate populations; binding resource unmeasured
Language transfers	Entropy predicts speech; symbolic costs transfer across genres	Known capacity account; physical cost and neutrality unidentified
Bacterial transfers	Population means and independent burden assays inform forecasts	Variability and fitted workload correction fail
Cantilever energy transfer	Some damping-to-energy forecasts transfer	Small average gain; spatial-loss and observation failures remain
Chemical / sediment constraints	Independent inputs improve transfer forecasts	Established rivals perform similarly or better; finite-accuracy limits
Chemostat nutrient pulse	Frozen two-budget reweighting fails	Constraint response needs explanation
Flare and lake geometry	Predicted exponents excluded or equivalence unresolved	Measurement and dependence affect interpretation
Seismicity	Qualification or estimator prerequisites fail	No real b–dimension comparison established

Table 1. Different studies answer different questions. Detailed results, all rival forecasts and observation limitations are retained in supporting material.

4.2 An allocation mechanism with an intervention

In a Python runtime that serializes execution, matrix-kernel workers compete for CPU time. Independently calibrated job costs predict both resource shares and inverse-cost job shares. Across 16 allocation trials, maximum CPU-share errors are 0.0035–0.0063 and job-share total-variation errors are 0.0021–0.0124. Replacing the largest-class worker with a second smallest-class worker predicts the allocation change within 0.0078. A linear-cost alternative ($q_j \propto k_j$) has job-share errors of 0.126–0.273 (intervention report; supporting S5.1).

The baseline uses logarithmic matrix-size classes; the intervention changes independently counted worker opportunities. This demonstrates a physical implementation of a conditional resource law in one engineered system.

4.3 Biological closure, predicted means and failed responses

Separately regrown *Dunaliella* cultures provide an intuitive inverse relation: a 10.4-fold difference in mean cell volume accompanies only a 2.5% difference in geometric mean total biovolume between large- and small-selected treatments. Frozen equal-biovolume forecasts beat fitted-size and equal-count alternatives in both held-out folds. Control capacity is 18–21% higher, and the limiting resource is unmeasured (Malerba et al., 2018; supporting S5.3). This is budget closure across separate populations, not log-orthopolic or linear-orthopolic allocation among coexisting sizes.

The cell-cycle mechanism predicts separate *E. coli* population means within 2.72% and 0.447%, but overpredicts variability and fails its joint criterion (Gangan and Athale, 2017; supporting S5.6). A close mean alone is insufficient.

A nutrient-pulse forecast fails in twelve held-out chemostat polycultures: none of the forecasts appreciably improves on persistence. A retrospective bound then shows that arbitrary C:N price changes still cannot explain 17/36 redistributions in eight vessels under transferred quotas and fixed opportunities (Wojcik et al., 2025; supporting S5.3/S5.10). That result rules out one proposed repair. Independent phosphorus assays also fail their count transfer; field partial-stock compatibility lacks allocation eligibility; and a fitted protein workload correction loses to direct transfer (supporting S5.11/S5.12/S5.16). These are species, cell or family comparisons, neither size-allocation measure.

4.4 Information costs and language transfer

Information per syllable supplies a cost in an established speech-capacity account. Seven Indo-European languages calibrate its information rate; written corpus estimates then predict speech duration in ten languages from eight other families. Language log MAE is 0.0741, lower than all four declared rivals, and all frozen criteria pass (Coupé et al., 2019; supporting S5.17). Individual errors can cancel within a language, and the comparison does not identify a metabolic resource or neutral word allocation.

For written word types, a letter/phoneme composite improves held-out genres over a letter-only power, but lexical frequencies perform better and cost shares remain uneven (supporting S5.5; Piantadosi et al., 2011). Both studies use specified counting measures, neither log-orthopolic nor linear-orthopolic size allocation. They assess the transfer of resource interpretations.

4.5 Geometric predictions and observation limits

Geometric resources give predictions that can fail. The primary aftershock productivity exponent, 0.63 with 95% interval $[0.51, 0.71]$, excludes the rupture-area and moment predictions; changing counting windows substantially changes the result. Flare ribbon density exponent 3.163, $[2.962, 3.457]$, excludes both proposed exponents, 2 and 7/3. Their registered rules retain inconclusive model-choice verdicts alongside these exclusions.

For lake radii, correcting physical geometry and spatial dependence removes the initial positive equivalence conclusion. The corrected survival exponent is 1.859, with basin-bootstrap interval $[1.785, 1.924]$: exact 2 is excluded, while the frozen equivalence band $[1.9, 2.1]$ remains overlapped. These are log-orthopolic resource hypotheses with declared coordinate transformations; complete comparisons are in supporting S5.7.

The seismic b–dimension test stops before a real hypothesis comparison: only six Southern California sequences qualify, against a required ten, and the Italian estimator prerequisite fails 45 of 56 required synthetic scenarios. These are observation boundaries, not evidence deciding between resource models.

4.6 Independent constraints and transport kinetics

Other transfers test independently specified constraints. Pure-gas calibration predicts chemical competition in 45 mixture conditions: shared-capacity Langmuir MAE is 0.247 mmol/g, versus 0.620 without competition and 0.243 for established IAST. The practical tolerance passes, but only 39/90 component predictions lie within reported uncertainties (Wedler et al., 2024; supporting S5.14).

Independent grain shape and wall corrections reduce error when transferring from narrow to wider channels, but separate old-material fits perform better and low-flux errors exceed fourfold (Deal et al., 2026; supporting S5.15). Earlier membrane pairs predict later chamber contrasts with error of 4.23% of each pair's initial total, close to empirical interpolation at 4.14% on the same scale; broad stock uncertainty prevents interpreting non-rejection as proof of conservation (Benstead et al., 2026; supporting S5.18). Species/vacancy counting, material-balanced counting and compartment volume define these respective measures, neither log-orthopolic nor linear-orthopolic size allocation. The transfers support specific physical accounts without uniquely favoring the general law over established theories.

5. Learning from validity, departures and failure

Normal aggregation and natural allocation offer complementary explanations. Gaussian normality can emerge from additive aggregation under central-limit conditions; we call the normalized neutral inverse-resource profile (2) a *natural (orthopolic) distribution*. Homogeneous geometric requirements yield power-law abundance under the declared log-orthopolic or linear-orthopolic measure. Both mechanisms can coexist: canonical harmonic modes have Gaussian coordinates and equal mean energies under mode-counting measure, neither log-orthopolic nor linear-orthopolic size allocation. Supporting S3.9 gives the conditions and distinguishes physical scaling relations from sampled distributions.

Uniform spatial cuts of a homogeneous rod conserve every realization's mass and give labeled fragments equal expected mass. Yet expected resource per logarithmic fragment mass is nonconstant. This analytic failure of log-orthopolic size-class equality follows before outcomes, without fitting a correction. Supporting S3.8 gives the finite-bin prediction and distinguishes this partition mechanism from exchange under class symmetry.

Departures have distinct explanations. Changing the measure or sampling alters the reported distribution; independently specified absorption, quantum levels or adsorption affinities predict physical constraints. Chemostat reweighting, bacterial variability and cantilever spatial-loss errors instead remain failed forecasts until a revised explanation succeeds on additional evidence.

Unavailable tests cannot decide between resource models, and unrestricted resource fitting supplies an identity (section 2.4). Further comparisons should change independently specified constraints so that rivals predict different allocation profiles, relaxation rates or cross-environment contrasts such as (4).

6. Cosmic implications and the proposed law

Stars in galaxies give a direct extension of the opening picture. Let k be stars per galaxy and $\phi(k) = dN_{\text{galaxies}}/dk$. Log-orthopolic allocation of stellar number predicts $k^2\phi(k) = C$: each decade of galaxy richness holds the same total number of stars. A fixed inventory also obeys $N_{\text{galaxies}}\bar{k} = N_{\text{stars}}$ exactly, but that accounting identity does not select the richness distribution.

Supporting S6 develops a **cosmological conjecture**: gravitational assembly can produce finite regimes of approximately equal mass per logarithmic host-mass interval. The conditional mechanism combines steady material current with equal residence time per logarithmic interval. Independently known current and drift would predict the mass-function normalization as well as its slope. Published galaxy, stellar and simulated subhalo distributions provide exploratory signs and departures. Stellar mass remains distinct from star number, and this allocation constant is not the cosmological constant Λ.

The natural-law proposal unifies a simple relation with several physical routes to it. Exact accounting establishes its mathematical core; specified symmetries and dynamics establish allocation realizations. Independently known constraints and restricted resource combinations produce further predictions. The retained empirical comparisons establish conditional successes, failures and observation boundaries. The next scientific task is to predict both where neutral allocation occurs and how it changes, using information beyond the distribution it explains.

Data and code availability

The repository and run registry retain sources, configurations, forecasts, algorithms and outcomes, including failures. The supporting material contains detailed derivations, complete comparisons and study reports. Source licences are in data/NOTICE.md. Published FIRAS reconstruction and bacterial figure summaries remain identified as such.

The make all target verifies inputs, runs tests and regenerates the original analyses; later studies have separate offline targets in their reports. The make paper target builds both manuscripts; registry-verify checks retained study references. The supplied 2017 manuscript was inspected privately and is not redistributed.

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